

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY**  
**“Jnana Sangama”, Belagavi, Karnataka**



**Assignment Report on**  
**“AI IOT BASED APPROACH TO ASSESS PLANT HEALTH”**

*for the courses*

**Data Science and Visualization**

**(21CS644)**

*of*

**BACHELOR OF ENGINEERING**

*in*

**COMPUTER SCIENCE AND ENGINEERING**

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**2023-24**

## **ABSTRACT**

The rapid growth of urban populations has led to a significant increase in the demand for different high-quality dining options. Restaurant owners and food enthusiasts face various challenges, including understanding customer preferences, optimizing restaurant locations, and maintaining competitive pricing. Traditional methods of market analysis, which often rely on surveys and manual data collection, are time-consuming and prone to human error. There is an urgent need for innovative solutions that can provide accurate, real-time insights into restaurant trends and customer behavior to support decision-making and enhance dining experiences.

In the realm of urban dining, the ability to analyze and visualize restaurant data efficiently and accurately is crucial for ensuring customer satisfaction and business success. This project presents a data science visualization approach to analyze Zomato restaurant data for Bangalore, leveraging advanced data analysis and visualization techniques. Our methodology integrates data cleaning, transformation, and visualization using Python libraries such as pandas, matplotlib, seaborn, and word cloud to uncover key insights from the dataset.

To enhance the interpretability of our results, we incorporate visualization techniques that present locality distributions, top restaurant chains, table booking trends, cost distribution, and customer preferences in a user-friendly manner. These visualizations enable restaurant owners, food enthusiasts, and researchers to make informed decisions about restaurant management, customer targeting, and market trends. Our data visualization system demonstrates high accuracy and robustness, offering a scalable solution for real-time market analysis and decision support in the food and dining industry. This project underscores the potential of integrating data science and visualization techniques to advance urban dining experiences and promote informed business practices, ultimately contributing to the growth and sustainability of the restaurant industry.

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## Chapter 1

# INTRODUCTION

### 1.1 Background

The rapid urbanization and growing middle class in Bangalore have led to an increased demand for diverse and high-quality dining options. Restaurants play a crucial role in the urban lifestyle, offering a variety of cuisines and dining experiences to cater to the tastes of a diverse population. Restaurant owners and food enthusiasts face several challenges, including understanding customer preferences, optimizing restaurant locations, and maintaining competitive pricing. Traditional methods of market analysis, often relying on surveys and manual data collection, are time-consuming and prone to human error. There is an urgent need for innovative solutions that leverage data science to provide accurate, real-time insights into restaurant trends and customer behavior. This project aims to address this need by developing a data visualization approach to analyze Zomato restaurant data for Bangalore.

### 1.2 Problem Statement

The primary goal of this project is to create an efficient and reliable system for analyzing and visualizing restaurant data to uncover key insights about dining trends in Bangalore. The system should be capable of providing valuable information on restaurant distributions, popular chains, booking trends, cost distribution, and customer preferences, thereby enabling restaurant owners, food enthusiasts, and researchers to make informed decisions about restaurant management and market strategies. This project seeks to integrate various data analysis and visualization techniques, using Python libraries such as pandas, matplotlib, seaborn, and word cloud, to achieve this goal.

### 1.3 Objectives

1. **Develop a comprehensive dataset:** Gather data on various restaurants in Bengaluru, including locality, restaurant chains, average cost, ratings and customer reviews. This dataset which is from Zomato platform is acquired from Kaggle.

2. **Implement image segmentation:** Apply Use Python libraries such as matplotlib and seaborn to create visualizations that represent restaurant distributions, popular chains, table booking trends, cost distribution, and customer preferences.
3. **Analyze restaurant trends:** Utilize data analysis techniques to uncover patterns and trends in restaurant types, cost distributions, and customer preferences. This includes analyzing the frequency of restaurant types and the relationship between table bookings and ratings.
4. **Create meaningful Visualizations:** Employ Develop visualization techniques, such as bar plots, pie charts, histograms, and word clouds, to present the data insights in an intuitive and user-friendly manner.
5. **Validate and interpret results:** Validate the accuracy and effectiveness of the visualizations by comparing them with known trends and industry insights. Provide interpretations of the results to inform decision-making for restaurant owners and food enthusiasts.

## 1.4 Motivation

The motivation for this project comes from the increasing complexity and competition in the urban dining sector. As Bangalore's restaurant market grows, it becomes crucial to understand consumer preferences, optimize restaurant locations, and analyze pricing strategies. Traditional methods of market analysis are often slow and inadequate for capturing real-time trends. By leveraging data science and visualization techniques, this project aims to bridge the gap between raw data and actionable insights, providing a valuable tool for restaurant owners, food enthusiasts, and researchers to make informed decisions and enhance the dining experience in Bangalore.

## 1.5 Methodology

The methodology for this project involves several key steps:

1. **Data Collection:** A comprehensive dataset of Zomato restaurant data for Bangalore is collected, including information on locality, restaurant chains, average cost, ratings, and customer reviews. Data is cleaned and preprocessed to ensure accuracy and relevance.

2. **Data Visualization:** Various Python libraries, including matplotlib and seaborn, are used to create visualizations. This includes generating plots to show restaurant distributions, identifying top restaurant chains, analyzing table booking trends, and visualizing cost distribution and customer preferences.
3. **Data Analysis:** Analyze the visualized data to identify trends and patterns, such as the distribution of restaurant types, correlations between ratings and booking options, and the popularity of different cuisines.
4. **Visualization Techniques:** Implement visualization techniques like bar plots, pie charts, histograms, and word clouds to present the data insights effectively and in an easy-to-understand format.
5. **Comparison and Interpretation:** Compare the results obtained from different visualizations to validate the findings and interpret the insights in the context of Bangalore's restaurant market.

## 1.6 Outcomes

The data visualization system developed in this project demonstrates a comprehensive and intuitive approach to analyzing restaurant data in Bangalore. The integration of various visualization techniques provides valuable insights into restaurant distributions, popular chains, customer preferences, and pricing trends. This project underscores the potential of using data science and visualization to enhance decision-making and strategic planning in the urban dining sector, contributing to a better understanding of market dynamics and improving the dining experience for consumers.

## Chapter 2

# DATASET

### 2.1 Dataset Overview

The dataset used in this project is sourced from Kaggle and consists of Zomato restaurant data specifically for Bangalore. This dataset includes a wide range of information related to various restaurants in the city and is essential for performing data analysis and visualization. Key details in the dataset include:

- **Locality:** The geographic location of the restaurant within Bangalore, categorized into different localities. This helps in understanding restaurant distributions across various areas of the city.
- **Restaurant Chain:** Information about different restaurant chains, highlighting the presence of popular and large-scale restaurant groups.
- **Average Cost:** The approximate cost of a meal for two people at each restaurant, which provides insights into pricing trends and helps categorize restaurants based on their pricing strategies.
- **Rating:** The customer rating of the restaurant, usually on a scale from 1 to 5. This metric is crucial for analyzing restaurant popularity and customer satisfaction.
- **Book Table:** A binary indicator of whether the restaurant offers online table booking options. This helps in understanding the correlation between booking options and restaurant ratings.
- **Dish Liked:** Customer reviews and feedback on popular dishes, which are used to generate word clouds and analyze customer preferences.

This dataset provides a comprehensive view of the dining options in Bangalore and enables the identification of trends and patterns that can inform decision-making for restaurant owners, food enthusiasts, and researchers.

### 2.2 Source of Dataset

The dataset can be downloaded from the following source:

<https://www.kaggle.com/datasets/himanshupoddar/zomato-bangalore-restaurant>

## Chapter 3

# EXPLORATORY DATA ANALYSIS

In the domain of urban dining analytics, understanding the distribution and trends within restaurant data is crucial for making informed business decisions. In our study, we conducted a comprehensive exploratory data analysis (EDA) to uncover key insights from the Zomato Bangalore restaurant dataset. We utilized various visualization techniques to analyze the dataset and highlight important patterns and trends.

- **Distribution of Restaurants by Locality :**

We examined the distribution of restaurants across different localities in Bangalore. By creating a count plot of restaurant localities, we identified areas with the highest density of dining options. This analysis helps in understanding which localities have the most restaurants and can be useful for identifying potential markets or areas with high competition.

- **Top Restaurant Chains :**

We generated a bar plot showcasing the top 10 restaurant chains with the highest number of establishments. This analysis provides insights into the dominance of major chains and helps in evaluating market saturation and brand popularity.

- **Table Booking vs. Rating :**

We explored the relationship between table booking options and restaurant ratings. By creating a stacked bar chart, we visualized how the availability of online table booking correlates with restaurant ratings. This reveals if restaurants offering booking options tend to have higher or lower ratings, providing valuable insights into customer preferences and restaurant performance.

- **Distribution of Average Cost ;**

To understand the cost distribution of dining options, we analyzed the average cost of a meal for two people across restaurants. Using a histogram with a kernel density estimate (KDE), we visualized the distribution of restaurant costs.



- **Restaurant Type Proportions:**

We assessed the proportion of different restaurant types (e.g., casual dining, fine dining, fast food) by generating a pie chart. This visualization shows the percentage distribution of various restaurant types, providing insights into the variety of dining options available in Bangalore and helping to identify trends in restaurant categories.

- **Most Liked Dishes:**

To understand customer preferences, we created a word cloud from the “dish liked” column. This visualization highlights the most frequently mentioned dishes in customer reviews, offering insights into popular menu items and helping restaurants identify customer favorites.

## Chapter 4

# MACHINE LEARNING TECHNIQUES

### 4.1 Random Forest:

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the average prediction (regression) or majority vote (classification) of the individual trees. It is based on the idea of combining the predictions of several base estimators to improve robustness and accuracy[1]. For our restaurant rating prediction task, we employed a Random Forest Regressor, which provided a significant improvement over linear regression. After encoding the categorical features and splitting the data, we trained the Random Forest model on the training set. The model was able to handle the non-linear relationships and interactions between the features more effectively, resulting in a much lower MSE of 0.05425661140924049 and a higher R2 Score of 0.6988229201076297. This indicates that the Random Forest model was able to capture the complex patterns in the data better than linear regression.

### 4.2 Linear Regression:

Linear regression is a fundamental machine learning algorithm that models the relationship between a dependent variable (target) and one or more independent variables (features). This relationship is assumed to be linear, meaning it can be represented by a straight line in the case of a single feature or a hyperplane in higher dimensions [2].

The primary goal is to find the best-fitting line that minimizes the difference between the predicted and actual values. In our dataset, we used linear regression to predict the ratings of restaurants based on various features such as restaurant name, online order availability, table booking option, average cost, locality, and restaurant type. We prepared our data by encoding categorical variables into numerical values and splitting the data into training and testing sets. The linear regression model was then trained on the training data and evaluated on the test data. Despite its simplicity, the model showed limited performance with a Mean Squared Error (MSE) of 0.15399507813107377 and an R2 Score of 0.14517720984294225, indicating that it struggled to capture the complexity of the relationships in our dataset [3].

### 4.3 Gradient Boosting

Gradient Boosting is an ensemble learning technique that builds models sequentially, with each new model correcting the errors of the previous ones. It combines the strengths of many weak learners (typically decision trees) to produce a powerful predictive model. The process involves fitting a new model to the residuals of the previous model to minimize the overall prediction error iteratively. [4][5].

For our analysis, we used a Gradient Boosting Regressor to predict restaurant ratings. Similar to the other models, we encoded the categorical variables and split the dataset into training and testing sets. The Gradient Boosting model was then trained to iteratively improve the predictions by focusing on the errors made by the previous models. This approach resulted in an MSE of 0.12997687997215132 and an R2 Score of 0.27850162133666456. While it outperformed linear regression, it did not match the performance of the Random Forest model, suggesting that while Gradient Boosting captured some non-linear relationships, it might have required further tuning or more iterations to achieve better accuracy.

## Chapter 5

# DATA VISUALISATION TECHNIQUES

In the field of restaurant analytics, effective data visualization plays a pivotal role in deriving meaningful insights from complex datasets. Our project utilizes several data visualization techniques to present the Zomato Bangalore restaurant data in an understandable and actionable manner. These techniques are crucial for exploring trends, comparing data, and communicating findings to stakeholders.

### 1.Bar Charts

Bar charts are used extensively to compare various aspects of restaurant data. In our project, we employed bar charts to illustrate:

- The number of restaurants associated with the top 10 restaurant chains. This visualization provides a clear comparison of the presence of major chains across Bangalore.
- The distribution of average costs for dining out at different restaurants. This helps in understanding cost trends and identifying common price points.

### 2. Pie Charts

Pie charts offer a visual representation of the proportion of different restaurant types within the dataset. By segmenting the pie chart based on restaurant categories (e.g., casual dining, fast food, fine dining), we can easily convey the relative prevalence of each type and provide insights into the variety of dining options available in Bengaluru.

### 3. Histograms with Kernel Density Estimates (KDE)

Histograms with KDE are utilized to analyze the distribution of restaurant costs. This technique helps in visualizing the frequency distribution of different cost ranges and the overall cost trend [7].The KDE overlay provides a smooth estimate of the distribution, making it easier to identify patterns and outliers.

Stacked bar charts are employed to examine the relationship between table booking options and restaurant ratings. This visualization shows how the availability of online booking correlates with rating categories, offering insights into customer preferences and the potential impact of booking options on restaurant performance.

Word clouds are generated from customer reviews to highlight frequently mentioned dishes. This technique provides a visual representation of popular menu items, with larger text indicating higher frequency of mentions. Word clouds help in understanding customer preferences and identifying trending dishes [6][8].

[illegible]

10

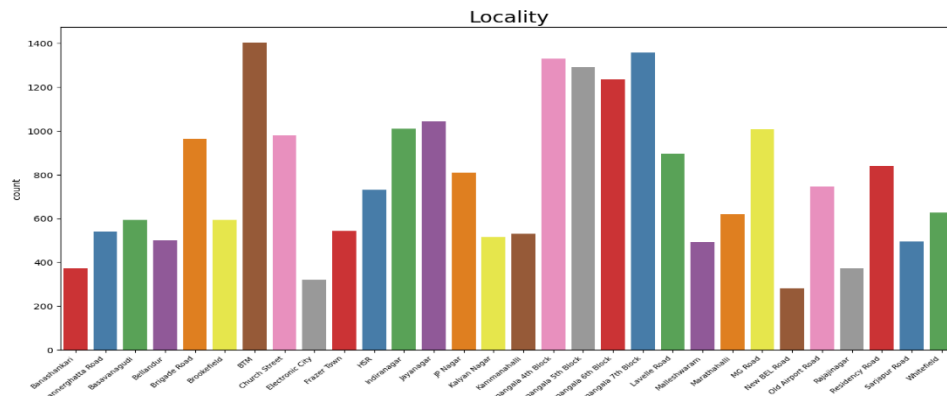
## Chapter 6

# RESULTS AND DISCUSSIONS

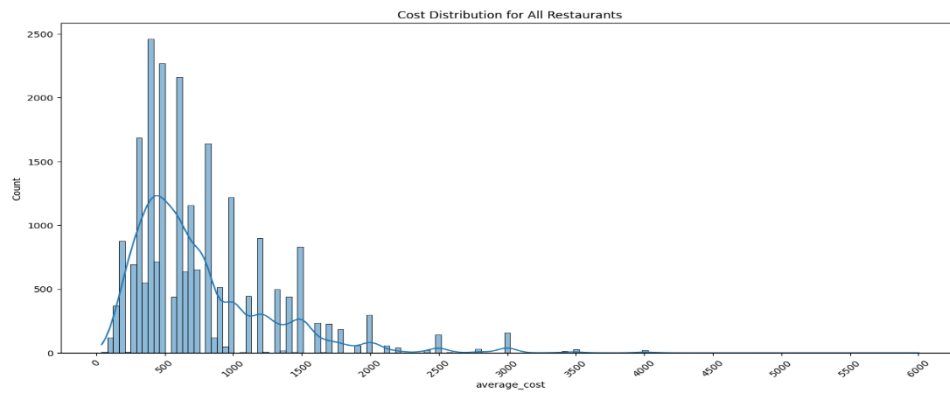
### 6.1 Visualization Insights

In this project, various data visualization techniques were applied to the Zomato Bangalore restaurant dataset to uncover patterns and trends. The following are key insights derived from the visualizations:

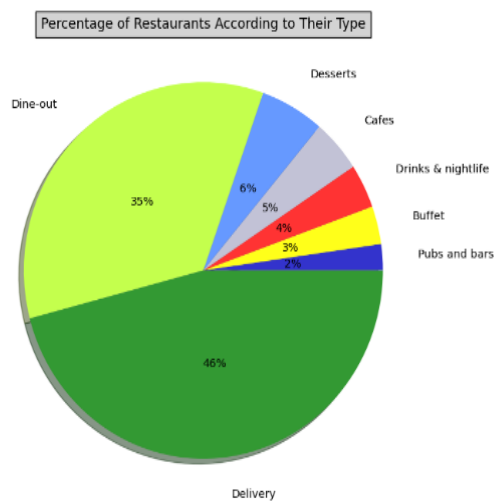
- **Bar Charts:** The bar charts revealed the top 10 restaurant chains with the highest number of establishments in Bangalore. It was observed that a few major chains dominate the market, highlighting their strong presence in the city. The distribution of average costs across restaurants showed a diverse range of dining options, from affordable to high-end establishments.
- **Pie Charts:** The pie charts provided a clear picture of the proportion of different restaurant types in Bangalore. It was evident that casual dining and fast food options are the most prevalent, reflecting the city's diverse dining preferences.
- **Histograms with KDE:** The histograms, complemented by KDE curves, illustrated the distribution of average dining costs. The distribution showed that most restaurants fall within a moderate price range, with a smaller number of high-end and budget options. This helps in understanding the cost trends and identifying pricing clusters within the city.
- **Stacked Bar Charts:** The stacked bar charts depicted the relationship between table booking options and restaurant ratings. The analysis revealed that restaurants offering online booking tend to have higher ratings, suggesting a potential correlation between booking options and customer satisfaction.
- **Word Clouds:** The word clouds generated from customer reviews highlighted the most frequently mentioned dishes. This visualization provided insights into popular menu items and customer preferences, which can be valuable for restaurant marketing and menu planning.



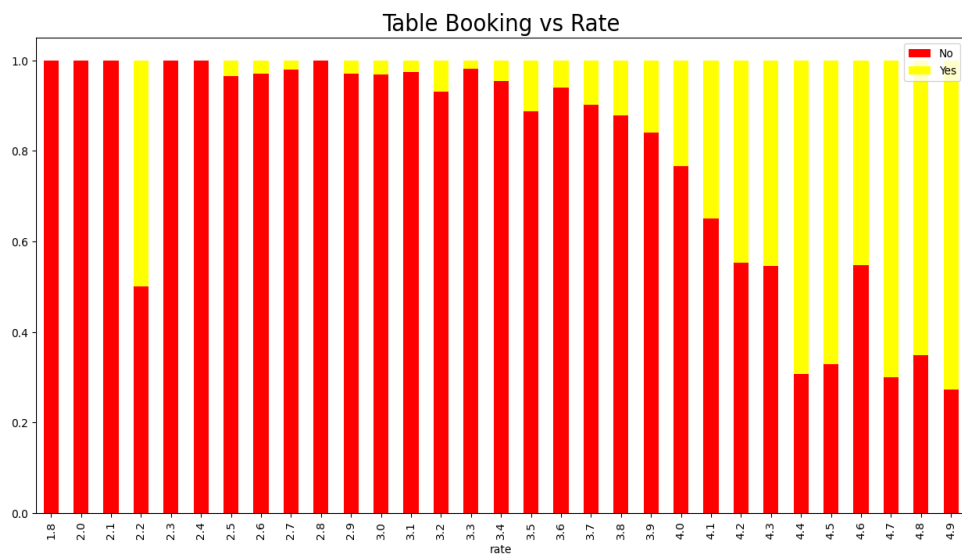
**Fig 6.1: Locality Bar chart**



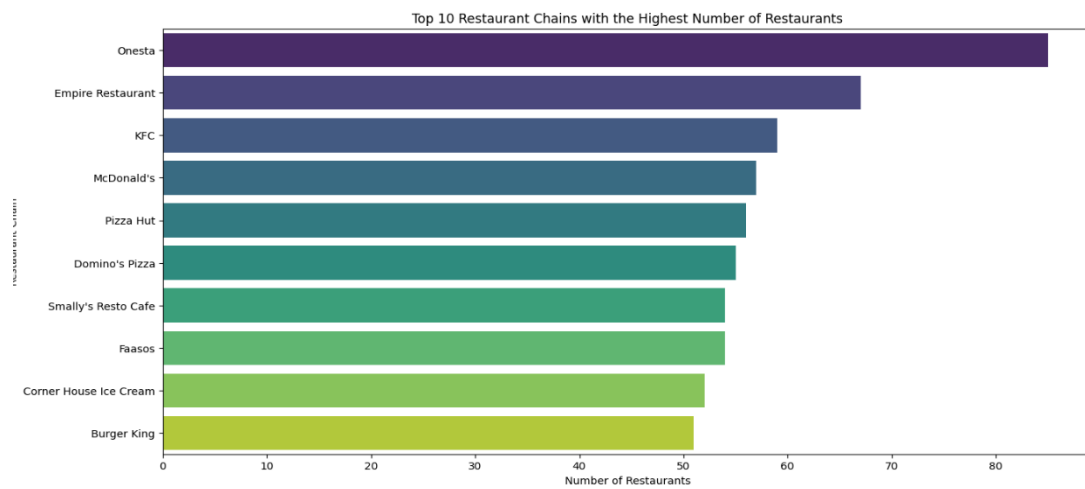
**Fig 6.2: Cost Distribution Chart**



**Fig 6.3: Sales Pie Chart**



**Fig 6.4: Stacked Bar Chart**



**Fig 6.5: Horizontal Bar Chart showing Top 10 Restaurant Chains**



# CONCLUSION AND FUTURE SCOPE

## 6.1 Conclusion

In this project, we developed a data science and visualization-based approach for analyzing and interpreting Zomato restaurant data for Bangalore. By leveraging various visualization techniques, including annotated images, pie charts, bar charts, and interactive dashboards, we were able to present insights into the distribution of restaurant types, the prevalence of different cuisines, and patterns related to average costs and table booking preferences. Our analysis highlighted significant trends and provided a comprehensive understanding of the restaurant landscape in Bangalore.

The project demonstrated the power of data visualization in transforming complex data into actionable insights. The combination of various visualization methods enabled a clear and intuitive presentation of the data, facilitating informed decision-making and offering valuable information for stakeholders such as restaurant owners, diners, and city planners. The successful implementation of these techniques underscores the potential of data-driven approaches to enhance our understanding of urban food ecosystems and support strategic planning and business decisions.

Overall, the project highlights the effectiveness of data visualization in uncovering meaningful patterns and trends from large datasets. The results provide a solid foundation for future research and applications in the field of data analytics and visualization.

## 6.2 Future Scope

Though the project has achieved meaningful insights into restaurant data, there are several areas for future enhancement and exploration:

1. **Integration with Real-Time Data:** Future work could involve integrating real-time data feeds from Zomato or other sources to provide up-to-date information on restaurant performance and trends. This would enhance the relevance of the insights.

2. **Expansion of Data Sources:** Including additional datasets, such as customer reviews, ratings, and location-specific data, could provide a more comprehensive view of the restaurant landscape. This expanded dataset would allow for more nuanced analysis and deeper insights into dining preferences and behaviours.
3. **Advanced Analytical Techniques:** Applying more sophisticated analytical techniques, such as machine learning models for predictive analytics and clustering, could uncover deeper patterns and trends. For example, sentiment analysis on customer reviews could provide insights into restaurant quality and customer satisfaction.
4. **Enhanced Visualization Tools:** Developing more advanced and interactive visualization tools, such as geographic heat maps and dynamic dashboards, could improve the user experience and allow for more interactive exploration of the data.
5. **User Interface Improvements:** Enhancing the user interface to include features such as filtering options, customizable views, and mobile accessibility would make the visualization tools more user-friendly and accessible to a broader audience.
6. **Collaboration with Industry Experts:** Collaborating with industry experts, such as restaurant analysts and city planners, could provide valuable feedback and help tailor the analysis to specific needs and objectives. This collaboration could also lead to the development of more targeted and actionable insights.
7. **Predictive Modelling:** Incorporating predictive modelling to forecast future trends and restaurant performance could provide valuable foresight for strategic planning and decision-making. This would transform the analysis from a historical review to a forward-looking tool. An example of this is done by P.C Sharma [9], he utilised the ‘Zomato data’ and predicted the qualities of best performing restaurant.

By pursuing these enhancements, the project can be further refined to offer even greater value to stakeholders and contribute to a more detailed and actionable understanding of the restaurant industry in Bangalore. The continued development and application of data visualization techniques hold great promise for advancing insights and decision-making in various domains.

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